Neural Network Practice Problems

1. For a square input of side $n = 32$, kernel $k = 5$, stride $s = 1$, padding $p = 2$, compute the output spatial size
$$n' = \left\lfloor \frac{n + 2p - k}{s} \right\rfloor + 1.$$

2. Same formula as above with $n = 28$, $k = 3$, $s = 2$, $p = 0$. Find n'.

3. A 3×3 filter has 4 input channels and 16 output filters (with bias). How many trainable parameters does this layer have?

4. A dense layer maps $F = 128$ features to $K = 10$ classes (with bias). Write P_{dense} and compute its value.

5. Explain why stacking many 3×3 filters can grow the receptive field. Give the receptive-field size after three stacked 3×3 convolutions with stride 1 and no dilation.

6. Write the residual update $\mathbf{y} = \mathbf{x} + f(\mathbf{x})$. Why does the identity path help training deep networks?

7. For $f(\mathbf{x}) = W_2\,\mathrm{ReLU}(W_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$ with matching dimensions, sketch one residual block and list its learnable parameters.

8. If channel dimensions do not match in a residual block, how do you adapt the skip connection? Write the projected form.

9. An activation map has size $H' \times W' \times C$. Write the global average pooling formula that yields a vector in $\mathbb{R}^C$.

10. After global average pooling to dimension $C = 64$ and a linear head with $K = 5$ classes (bias included), how many head parameters are there?

11. Contrast flatten-then-dense with global average pooling: give two benefits of pooling and one drawback.

12. A 4×4 single-channel input is convolved with a 2×2 filter, stride 1, no padding. What is the output spatial size?

13. Given input patch $X = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and filter $W = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, bias $b = 0$, compute the scalar convolution result, then apply ReLU.

14. For the same X and $W = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $b = -2$, compute $\mathrm{ReLU}(W \cdot X + b)$.

15. Max-pool a 2×2 window $\begin{bmatrix} 1 & 5 \\ 3 & 2 \end{bmatrix}$. What is the pooled value?

16. Average-pool the same window. What is the pooled value?

17. A network processes a 28×28 grayscale image through Conv+ReLU $\to$ MaxPool $\to$ Flatten $\to$ Dense $\to$ Softmax. List each stage's role.

18. For $m = 8$ filters of size 5×5 with $k = 1$ input channel (no bias), compute $P_{\text{conv}} = m \cdot k \cdot 5 \cdot 5$.

19. Same layer with bias included: add the bias count and give the new total.

20. Explain translation equivariance of convolution in one or two sentences.

21. Why is weight sharing in convolution parameter-efficient compared with a fully connected layer on the same image?

22. An input is $64 \times 64 \times 3$. After one 3×3 convolution with 32 filters, same padding, stride 1, what is the output tensor shape?

23. Then apply 2×2 max-pooling with stride 2. What is the new spatial size?

24. Write a one-line comparison: local filters vs. global interactions across positions.

25. If a model needs long-range dependencies early, why may deep stacks of small kernels be inefficient?

26. State the evidence lower bound (ELBO) for a latent variable model with encoder $q_\phi(z \mid x)$ and decoder $p_\theta(x \mid z)$.

27. Name the two terms in the ELBO and state what each encourages during training.

28. Write the reparameterization
$$z = \mu + \sigma \odot \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I).$$
Why is this needed for gradient-based training?

29. For a diagonal Gaussian q and prior $p(z) = \mathcal{N}(0, I)$, write the closed-form KL
$$D_{\text{KL}} = \tfrac{1}{2} \sum_{i=1}^{d} (\mu_i^2 + \sigma_i^2 - 1 - \log \sigma_i^2).$$

30. Compute D_{KL} for $\mu = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\sigma = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

31. Compute D_{KL} for $\mu = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\sigma = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

32. Compute D_{KL} for $\mu = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $\sigma = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

33. Compute D_{KL} for $\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\sigma = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. What does this value tell you?

34. Compute D_{KL} for $\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\sigma = \begin{bmatrix} e \\ 1 \end{bmatrix}$.

35. If $\sigma_i \to 0$ for some i while $\mu_i \neq 0$, qualitatively what happens to the KL term?

36. Explain reconstruction loss for a Gaussian decoder: how does $\|\mathbf{x} - \boldsymbol{\mu}_\theta(z)\|^2$ appear?

37. Write a Monte-Carlo estimate of $\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)]$ using one sample z.

38. Training minimizes the negative ELBO. Write the loss as reconstruction term + KL term (signs included).

39. If the KL weight is set to zero, what failure mode can appear in the latent codes?

40. If the KL weight is very large, what failure mode can appear in reconstructions?

41. Given $\mu = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix}$ and $\sigma = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, sample conceptually with $\varepsilon = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$: compute z.

42. Same μ, σ with $\varepsilon = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$: compute z.

43. For $d = 2$, expand D_{KL} fully as a sum of two scalar terms (no $\sum$ symbol).

44. Show that when $\mu = \mathbf{0}$ and $\sigma = \mathbf{1}$, $D_{\mathrm{KL}} = 0$.

45. True or false: the reparameterization moves randomness into ε so μ and σ remain differentiable paths. Justify.

46. A latent code is $z \in \mathbb{R}^{16}$. Encoder outputs $\mu, \log \sigma^2 \in \mathbb{R}^{16}$. How many scalars does the encoder emit?

47. Why do practitioners often predict $\log \sigma^2$ instead of σ directly?

48. Briefly contrast an autoencoder (deterministic bottleneck) with a model that samples $z \sim q_\phi(z \mid x)$.

49. Write one sentence on how the prior $p(z) = \mathcal{N}(0, I)$ acts as a regularizer.

50. For $\mu = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\sigma = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, compute D_{KL}.

51. Write scaled-dot-product attention

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}}\Big)V.$$

State the roles of Q, K, and V.

52. Why divide by $\sqrt{d_k}$ before the softmax?

53. For $d_k = 2$,

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Compute $QK^\top$ and $S = QK^\top/\sqrt{2}$.

54. Using S from the previous problem, write the row-wise softmax matrix A in exact form (exponentials allowed).

55. With $V = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and your A, write the expression for AV (exact form is fine).

56. Compute $QK^\top$ for

$$Q = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

57. For that $QK^\top$, form $S = QK^\top/\sqrt{2}$. Write S explicitly.

58. If $V = I_2$, what is the relation between the attention output and the softmax matrix A?

59. Compute $QK^\top$ for
$$Q = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

60. Show that $S = QK^\top/\sqrt{2}$ equals $\mathrm{diag}(\sqrt{2}, \sqrt{2})$ for the previous Q, K.

61. For
$$Q = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$$
compute $K^\top$ and $QK^\top$.

62. With $V = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, explain why the final attention output equals $2A$ if $A = \mathrm{softmax}(S)$.

63. Write multi-head attention:
$$\mathrm{head}_i = \mathrm{Attention}(QW_i^Q, KW_i^K, VW_i^V), \quad \mathrm{MHA} = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)W^O.$$

64. Why can multiple heads capture different relation types among positions?

65. Write a residual block update
$$\mathbf{x}' = \mathrm{LayerNorm}(\mathbf{x} + \mathrm{SubLayer}(\mathbf{x})).$$
Name two common choices of SubLayer.

66. State one benefit of residual connections and one benefit of LayerNorm in deep stacks.

67. Compare computational parallelism of attention over a length-T sequence versus a sequential recurrent step-by-step update.

68. For two tokens with scores $s = [0, 0]$, compute $\mathrm{softmax}(s)$.

69. For scores $s = [2, 0]$, write $\mathrm{softmax}(s)$ in exact form.

70. For scores $s = [\ln 2, \ln 2]$, compute $\mathrm{softmax}(s)$.

71. If one attention score is much larger than the others, describe the effect on the softmax weights.

72. A sequence has length $T = 4$ and model width $d = 8$. What is the shape of Q if each token is projected to $d_k = 8$?

73. For single-head attention, the score matrix $QK^\top$ has shape ? when there are T tokens.

74. Explain masking in causal (autoregressive) attention in one or two sentences.

75. Position encodings are added to token embeddings. Why are they needed if attention itself has no inherent order?

76. Write the cost intuition: attention score computation scales how with sequence length T (big-O in T), for fixed width?

77. An encoder produces feature maps of size $14 \times 14 \times 128$. After global average pooling and a $K = 10$ linear head (with bias), compute the head parameter count.

78. A 1×1 convolution with $C_{\text{in}} = 64$, $C_{\text{out}} = 32$, and bias: how many parameters?

79. Same-padding 3×3 convolution keeps spatial size. For input 56×56, what is the output spatial size?

80. Valid (no) padding 3×3 convolution, stride 1, input 56×56: output spatial size?

81. Two residual blocks in sequence: write $\mathbf{x}_2 = \mathbf{x}_1 + f_2(\mathbf{x}_1)$ with $\mathbf{x}_1 = \mathbf{x}_0 + f_1(\mathbf{x}_0)$. Expand $\mathbf{x}_2$ in terms of $\mathbf{x}_0$.

82. For ELBO training, list three quantities the encoder must provide (or imply) to form z and the KL term.

83. Given $\mu = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$, $\sigma = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, compute D_{KL}.

84. Given $\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\sigma = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, compute D_{KL}.

85. A mini-batch has B examples. Using one sample $z^{(b)}$ per example, write a batch estimate of the average reconstruction term.

86. Explain in one sentence how sampling z enables generating new $\hat{x}$ from the decoder by drawing $z \sim p(z)$.

87. Compute $QK^\top$ for
$$Q = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

88. For
$$Q = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad K = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$
compute $QK^\top$.

89. Using $d_k = 1$ (scalar case) with $q = 2$, $k = 1$, $v = 4$, compute attention output $\text{softmax}(qk/\sqrt{1})\, v$ carefully treating softmax on a single logit.

90. For two keys, scores $s = [1, -1]$, values $v_1 = 0$, $v_2 = 2$, compute the attention output scalar.

91. Show steps for scores $s = [0, 1]$, values $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$: find the output vector.

92. Write LayerNorm residual for attention then FFN as two successive updates (formulas only).

93. A model uses $h = 4$ heads with $d_{\text{model}} = 64$. What is a typical per-head d_k if $d_{\text{model}} = h\, d_k$?

94. If $W^Q \in \mathbb{R}^{d \times d_k}$, input $X \in \mathbb{R}^{T \times d}$, what is the shape of XW^Q?

95. Name one advantage of global pairwise interactions and one limitation relative to local convolution on images.

96. A classifier head on a pooled vector $f \in \mathbb{R}^C$ uses softmax. Write $\hat{y} = \text{softmax}(Wf + b)$ and the dimensions of W for K classes.

97. For convolution output size, show that $n = 7$, $k = 3$, $s = 2$, $p = 1$ gives $n' = 4$.

98. Design a tiny numerical check: pick any 2×2 $Q = K = V = I$, compute attention output, and state what you expect qualitatively.

99. Using KL formula, prove that each dimension contributes independently (no cross terms).

100. If $\sigma_i = 1$ for all i, simplify D_{KL} to an expression involving only μ.

101. If $\mu = \mathbf{0}$, simplify D_{KL} to an expression involving only σ.

102. Compute attention scores $S = QK^\top/\sqrt{2}$ for

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and comment which token the first query prefers.

103. A pipeline ends with either a CLS-style summary token or global average pooling before classification. State one similarity and one difference in how the summary vector is formed.

104. Write a compact checklist for a full attention forward pass: (1) form Q, K, V, (2) scores, (3) softmax, (4) weight V, (5) optional output projection.

Formula sheet (reference)

$$\begin{aligned} n' &= \left\lfloor \frac{n + 2p - k}{s} \right\rfloor + 1, \\ P_{\mathrm{conv}} &= m \cdot k \cdot k_h \cdot k_w \quad (+m \text{ if bias}), \\ P_{\mathrm{dense}} &= K(F+1), \\ z &= \mu + \sigma \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \\ D_{\mathrm{KL}} &= \tfrac{1}{2} \sum_i (\mu_i^2 + \sigma_i^2 - 1 - \log \sigma_i^2), \\ \mathrm{Attention}(Q, K, V) &= \mathrm{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}}\Big)V. \end{aligned}$$